

Title: Understanding the Composition of Sub-Neptunes with JWST: The First Constraints on the Internal Structure of the Coolest Sub-Neptune To Date

Abstract:

Sub-Neptune exoplanets are the most common type of exoplanet in the solar neighborhood, yet no analog exists in our Solar System. Consequently, the range of plausible internal structures for sub-Neptunes is poorly constrained, thus motivating JWST observations of sub-Neptunes. However, JWST is currently missing an important parameter space: cool sub-Neptunes. All but one of the sub-Neptunes observed by JWST so far have equilibrium temperatures above 310 K, almost 60 K hotter than Earth's equilibrium temperature. A planet's equilibrium temperature can have a major impact on its structure, for example influencing whether a planet is able to retain an atmosphere or an ocean. Without observing cool sub-Neptunes, we cannot draw conclusions about whether a single structural model could explain all sub-Neptune interior structures, or whether different models are needed depending on a planet's equilibrium temperature. We therefore propose to conduct JWST/NIRISS and NIRSpec transit spectroscopy observations of TOI-2257 b, a sub-Neptune exoplanet with an equilibrium temperature of 256 K (lower than any sub-Neptune yet observed by JWST). This program will enable atmospheric retrievals that will constrain the internal structure of TOI-2257 b. With a relatively short orbital period despite its low equilibrium temperature, TOI-2257 b uniquely balances observability with scientific value, making it the ideal candidate to launch investigations of cool sub-Neptune structures. Given the ubiquity of sub-Neptune exoplanets, this program will provide planetary composition information that is critical to answer broader questions about planetary formation and evolution.

- **Scientific Justification**

Sub-Neptunes Are Common, But Mysterious

Sub-Neptune exoplanets—defined as exoplanets between 2-4 Earth radii—are the most common type of exoplanet in our Solar Neighborhood [4], yet there is no analog for this type of planet in our Solar System. Thus, the range of plausible internal structures for sub-Neptunes is poorly constrained. Many theoretical structures have been proposed to explain the bulk properties of sub-Neptunes [e.g., 5], but because we lack an up-close example to validate these models, we require observations of sub-Neptune exoplanets to sufficiently investigate their internal structures.

The campaign to observe and uncover sub-Neptune internal structures has already begun thanks to the cutting-edge capabilities enabled by JWST. However, these observations have not yet covered an important parameter space: cool sub-Neptunes. Thus far, all but one of the sub-Neptunes observed by JWST have equilibrium temperatures below 310 K [6]. This is important because equilibrium temperature (a measure of the amount of radiation a planet receives from its star) can have a major impact on planet structure [3]. Without observing cool sub-Neptunes, we will be unable to determine whether one model can describe all sub-Neptune structures, or whether we need different models for planets that receive different levels of irradiation from their host stars. The ambiguity of sub-Neptune structures has already sparked a debate in the field prompting many to draw conclusions about which structures are possible for sub-Neptune exoplanets [e.g., 1, 2], so it is vital that we fill this gap in sub-Neptune observations now.

We propose to conduct JWST/NIRSpec and NIRISS transit spectroscopy observations of TOI-2257 b, a sub-Neptune exoplanet with an equilibrium temperature of 256 K. This is lower than the equilibrium temperature of any other sub-Neptune that has yet been observed by JWST. JWST is the only facility with a high enough spectral resolution to perform detailed characterization of sub-Neptune-sized exoplanets like TOI-2257 b. We will use atmospheric retrievals of the spectra of this sub-Neptune to place constraints on its internal structure, thus launching an investigation of cool sub-Neptune structures. Ultimately, this investigation will uncover information about planetary composition that is necessary for studies of planetary formation and evolution.

Existing JWST Observations Lack Cool Sub-Neptune Representation

Equilibrium temperature, a measure of the amount of radiation a planet receives from its star, can have a major impact on planetary internal structure [e.g., 3]. However, all but one of the sub-Neptunes observed by JWST thus far have equilibrium temperatures higher than 310 K (almost 60 K hotter than Earth’s equilibrium temperature) [6]. Thus, existing observations are not fully exploring the impact of equilibrium temperature on internal structure. These impacts have been extensively studied in Jovian planets; for example, it has been shown that a higher equilibrium temperature can inflate the radii of Jovian planets and push their radiative zones deeper [7]. Since this correlation exists for larger planets, it is possible that one may also exist for smaller planets. In order to investigate that possibility, we need observations of a range of sub-Neptunes with varying equilibrium temperatures. Because JWST has so far observed mostly warm and hot sub-Neptunes, we are missing key observations necessary to assess the impact of equilibrium temperature on sub-Neptune internal structure.

Little is known about sub-Neptune exoplanets, so investigating their structural dependence on equilibrium temperature via atmospheric retrievals of cool sub-Neptunes will be a major step forward in better understanding this class of exoplanets. This program will conduct transit spectroscopy observations of TOI-2257 b, a sub-Neptune with a cooler equilibrium temperature than any sub-Neptune exoplanet yet observed by JWST. These observations will reveal whether and how cool sub-Neptune structures differ from those of warm sub-Neptunes.

This Proposal: Atmospheric Retrieval of TOI-2257 b, Coolest Sub-Neptune Yet

This program will observe the exoplanet TOI-2257 b. This is a $2.2R_{\oplus}$ exoplanet with an equilibrium temperature of 256 K [8], which would make it the coolest sub-Neptune to be observed by JWST to date. **Figure 1** depicts the properties of TOI-2257 b compared to sub-Neptunes that have been observed by JWST, highlighting its cool temperature. The fact that most sub-Neptunes observed by JWST thus far are warmer is not surprising; exoplanets with higher equilibrium temperatures are typically easier to observe with the transit method because they orbit closer to their host star for a given star type and therefore transit more frequently. However, TOI-2257 b orbits a cool M dwarf star and therefore has a low equilibrium temperature despite a relatively short orbital period of 35.19 days [8]. This is short enough to observe multiple transits of this planet in one program. JWST is the only instrument that can perform these observations because it is space-based, eliminating the obstacle of telluric absorption that would impede atmospheric retrievals of TOI-2257 b; and because it is the only facility with high enough spectral resolution and sensitivity to probe the atmospheres of smaller exoplanets. With a JWST Transmission Spectroscopy Metric of 32.71 [8], TOI-2257 b is a strong candidate for JWST characterization that will enable key investigations into sub-Neptune structures.

We propose to collect transmission spectra during transits of TOI-2257 b with JWST/NIRISS and NIRSpec G395H. We propose the collection of two transits to increase the signal to noise ratio of the data. We will run atmospheric retrieval models on the resulting data to constrain the composition of the atmosphere of TOI-2257 b. Using this composition information, we will draw inferences about the structure of this planet; for example, a strong detection of water may indicate that the planet has a liquid water ocean, while a detection of volcanic gases may indicate that it has a magma ocean. As shown in **Figure 1**, these inferences will provide the first information about such a cool sub-Neptune and will inform how the structures of cooler sub-Neptunes may differ from those of the warmer sub-Neptunes we have already observed.

The results of atmospheric retrievals on the transmission spectra generated by this program will place constraints on the internal structure of TOI-2257 b, the coolest sub-Neptune that would be observed by JWST to date. We will use these constraints to determine the likelihood that different models are needed to describe the structures of cool sub-Neptunes compared to warmer sub-Neptunes.

Knowledge of Planetary Structures Informs Investigations of Planetary Evolution and Habitability

Constraints on the internal structure of TOI-2257 b will inform whether we need different models to explain the structures of sub-Neptunes with different equilibrium temperatures. More than one cool sub-Neptune will need to be observed to fully answer

Properties of Sub-Neptunes with Atmospheres Characterized by JWST Compared to TOI-2257 b

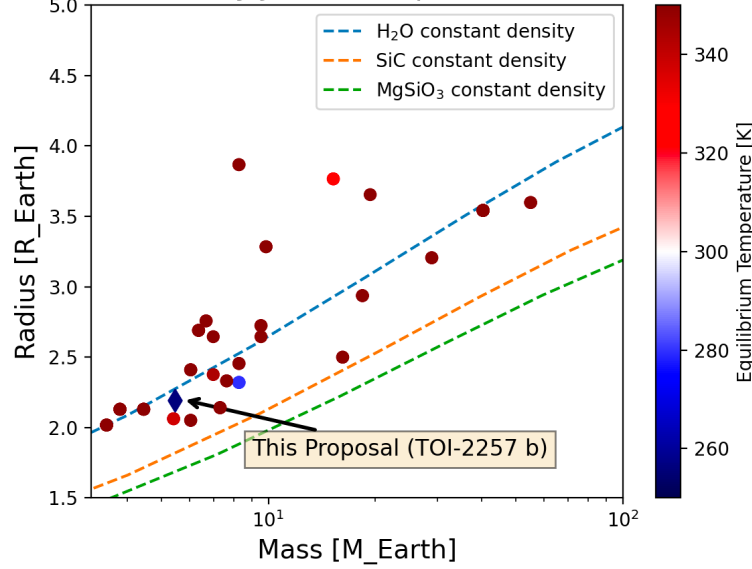


Figure 1: JWST transit spectroscopy of TOI-2257 b will characterize the atmosphere of the sub-Neptune with the lowest equilibrium temperature yet observed by JWST. Plotted are all planets with radii between 2-4 Earth radii (sub-Neptune sized) that have been observed by JWST, color-coded by equilibrium temperature. Isodensity curves for compounds that are often important elements of planetary composition are plotted to demonstrate that sub-Neptunes have varying densities and potentially varying compositions. Observations are needed to investigate the nature and diversity of sub-Neptune composition and internal structure. However, most of the sub-Neptunes observed by JWST thus far are hot; all but one have equilibrium temperatures above 310 K. The target of this program, TOI-2257 b (plotted as a diamond), has a lower equilibrium temperature (256 K) than any of the sub-Neptunes characterized by JWST. Thus, this program will probe the dependence of sub-Neptune interior structures on equilibrium temperature more extensively than any other observations to date. These observations will inform whether we need different theoretical models to describe the interior structures of sub-Neptunes with different equilibrium temperatures. This figure was developed by the authors for this proposal; JWST observation data is from the TrExoLiSTS database [6].

this question, but TOI-2257 b best optimizes the balance between cool equilibrium temperature and high observability and as such is the ideal candidate to launch these investigations as we begin to discover more cool sub-Neptunes.

It is vital that we conduct this study now because JWST is finally enabling our ability to explore the structures of smaller exoplanets and we are beginning to develop detailed models of these planets [e.g. 1, 2], but we have yet to determine whether and how those models should account for equilibrium temperature. This program relates to the key JWST science question: “From our own solar system to distant star systems - what can we understand about planet formation, evolution, and the suitability of planets as habitats?” Planetary structure is a result of a planet’s formation and evolution, so learning how structure is impacted by equilibrium temperature will inform conclusions about how planet formation and evolution are impacted by equilibrium temperature. This will advance not only sub-Neptune studies, but studies of the formation and evolution of all classes of planets, since formation and evolution are inherently interconnected across planet types.

References

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